

Our cat, Gado, has been kidnapped. The kidnapper has sent us a document with their requests in MS Word Document format. We have converted the document to PDF format and extracted the image from the MS Word file for your convenience.

STEP 1: Changing the directory and listing the files in that directory

```
root@ip-10-66-81-82:~# cd /root/Rooms/introdigitalforensics
root@ip-10-66-81-82:~/Rooms/introdigitalforensics#
root@ip-10-66-81-82:~/Rooms/introdigitalforensics# ls
letter-image.jpg  ransom-letter.doc  ransom-letter.pdf  ransom-lettter-2.zip
```

After listing all the files, I was prompted to look in the ransom pdf file.

STEP 2: Using *pdftinfo* which displays various metadata related to a PDF file, such as title, subject, author, creator, and creation date.

```
root@ip-10-66-81-82:~/Rooms/introdigitalforensics# pdftinfo ransom-letter.pdf
Title:          Pay NOW
Subject:        We Have Gato
Author:         Ann Gree Shepherd
Creator:        Microsoft® Word 2016
Producer:       Microsoft® Word 2016
CreationDate:   Wed Feb 23 09:10:36 2022 GMT
ModDate:        Wed Feb 23 09:10:36 2022 GMT
Tagged:         yes
UserProperties: no
Suspects:       no
Form:           none
JavaScript:     no
Pages:          1
```

I was able to find the author – Ann Gree Shepherd.

Next, the goal was finding the location of where the image was taken using the image file in the current directory.

STEP 3: I was prompted to use *exiftool* used to read and write metadata in various file types, such as JPEG images, to find the street name.

```

root@ip-10-66-81-82:~/Rooms/introdigitalforensics# ls
letter-image.jpg  ransom-letter.doc  ransom-letter.pdf  ransom-lettter-2.zip
root@ip-10-66-81-82:~/Rooms/introdigitalforensics# exiftool letter-image.jpg
ExifTool Version Number      : 11.88
File Name                    : letter-image.jpg
Directory                   : .
File Size                   : 124 kB
File Modification Date/Time  : 2022:02:23 08:53:33+00:00
File Access Date/Time       : 2022:02:23 09:12:00+00:00
File Inode Change Date/Time  : 2022:03:04 12:15:19+00:00
File Permissions             : rwxr-xr-x

```

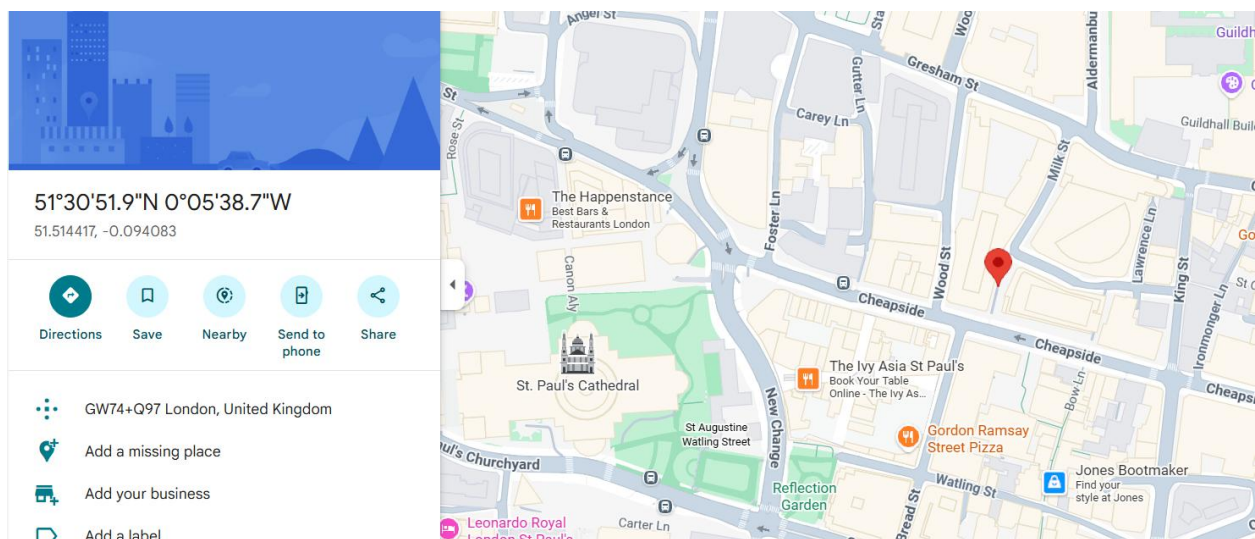
I found the coordinates when I scrolled a bit further down.

```

Thumbnail Image              : (Binary data 4941 bytes, use -b option to extract)
GPS Latitude                 : 51 deg 30' 51.90" N
GPS Longitude                : 0 deg 5' 38.73" W

```

I then took the coordinates and used them on Google Maps and found the exact location which I then used to find the street name – Milk St



Summary Conclusion

In this digital forensics exercise, I analysed files within a directory to extract key evidence. By reviewing PDF metadata with *pdfinfo*, I identified the author as **Ann**

Gree Shepherd. I then examined an image using *exiftool*, which revealed embedded GPS coordinates. After mapping the coordinates on Google Maps, I successfully determined the location and identified the street name as **Milk St.** This process demonstrated the effective use of metadata tools to uncover important forensic details.